WyreStorm

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Prepared by: Solutions Team
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Northstar Operations Room

WYRESTORM | TECHNICAL PROPOSAL

Executive Summary

Route two operator sources to four displays with resilient 4K distribution.

This document sets out the WyreStorm recommendation for this requirement, how it will work in use, the equipment involved, and what remains to be confirmed before it is issued as a final customer proposal.

Room at a glance

Operations room

Route two operator sources to four displays with resilient 4K distribution.

Recorded room and design details

Room dimensions, capacity and site conditions require confirmation.

What this space delivers

- **Operational continuity:** Operators retain controlled access to approved sources.

In this room

Grouped by the recorded equipment role. Item references match the equipment schedule; optional items remain subject to selection. Third-party equipment and services are detailed in Complete System Design and Responsibilities.

Encode operator sources

ITEM	MODEL	DESCRIPTION	QTY	SELECTION
1	NHD-600-TX	NetworkHD 600 encoder	2	Required

Decode display feeds

ITEM	MODEL	DESCRIPTION	QTY	SELECTION
2	NHD-600-RX	NetworkHD 600 decoder	4	Required

Routing control

ITEM	MODEL	DESCRIPTION	QTY	SELECTION
3	NHD-CTL-PRO-V2	NetworkHD controller	1	Required

Solution Overview

NetworkHD

What successful delivery looks like

- Every approved source-to-display route passes functional testing.

Customer Requirement

Requirement to be agreed: Route two operator sources to four displays with resilient 4K distribution.

Discovery Conversation

The recommendation in this document is based on the discovery conversation captured below: the question asked, the closest governed answer, and the customer's own wording where recorded. Rows marked "Confirmed with customer" were verified with the customer during discovery; rows marked "To be confirmed" are still open and must be verified before final design sign-off. Nothing here is presented as a settled fact that was not said during discovery.

QUESTION ASKED	GOVERNED ANSWER	CUSTOMER WORDING	STATUS	CAPTURE CONFIDENCE
How many displays must receive independently routed content?	Four displays	Each operator needs independent routing.	Confirmed with customer	High confidence (12)

The "Capture confidence" column shows how strongly the customer's wording matched the closest governed answer: "High confidence" is a strong exact match, "Matched" is a curated-phrase match, and "Low confidence — verify before quote" is a partial match that must be re-verified with the customer before quoting. The number in brackets is the match score behind the tier.

Where a row is a note-only capture, the answer was still open at the time of writing and must be confirmed before final design sign-off.

Recommended Solution

Recommendation: NetworkHD is the leading product-family path for this requirement, based on the recommendation evidence gathered so far (family confidence 96/100).

The confirmed routing and resilience requirement needs a scalable 10GbE fabric.

Information required: Confirm redundant power and switch failover behaviour during commissioning.

Required WyreStorm products: NHD-600-TX - NetworkHD 600 encoder; NHD-600-RX - NetworkHD 600 decoder; NHD-CTL-PRO-V2 - NetworkHD controller

Practical Operation

This is what the system will do for the people using the room, day to day.

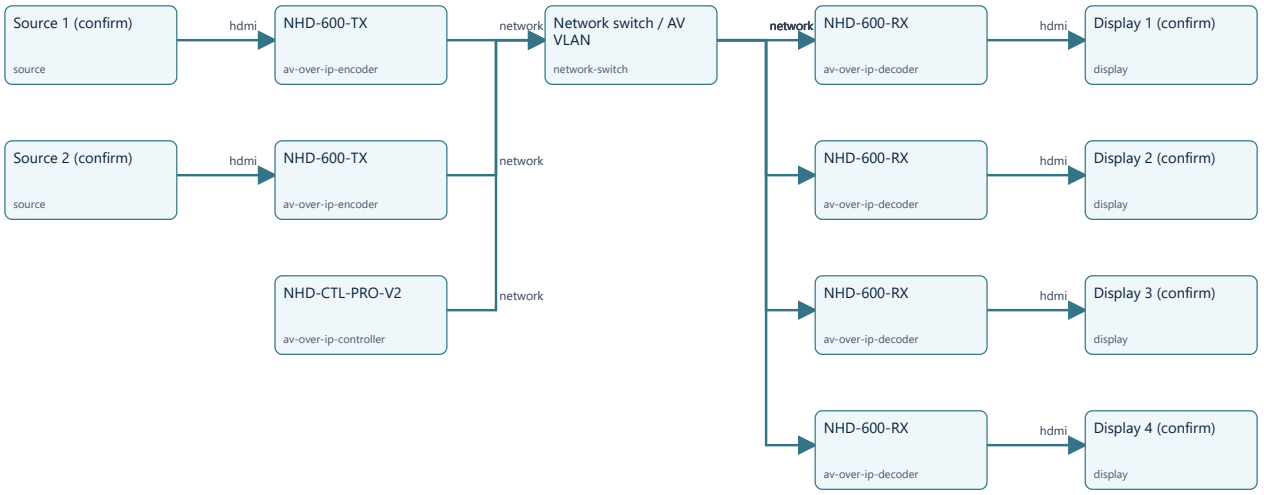
1. Operators select an approved source and route it to any display.

Commercial and Operational Benefits

- A consistent, repeatable connection experience across the room(s) covered by this proposal.
- AV-over-IP provides a potential expansion path, subject to compatible endpoints, network capacity and the final design.
- Final commercial and operational benefit statements should be confirmed against the customer's specific business or KPI outcome before proposal issue.

Technical Architecture

The following diagram and cable schedule explain how the recommended solution connects together, derived directly from the equipment schedule below.



Concept drawing: confirm physical connections and site interfaces before installation.

- Transport is shown generically because signal distance or connector evidence is not confirmed.
- Connections without a verified interface profile are proposed topology only. Confirm named ports, endpoint pairing, power and cable types before installation.
- Control routing requires design: Specify the controller and controlled-device interfaces before allocating control cables.
- Cable length not confirmed for hdmi run: The local source to encoder connection uses hdmi transport (max 15m). Confirm the actual cable run length is within spec before quoting.
- Cable length not confirmed for hdmi run: The local source to encoder connection uses hdmi transport (max 15m). Confirm the actual cable run length is within spec before quoting.
- Cable length not confirmed for hdmi run: The Decoder to display connection uses hdmi transport (max 15m). Confirm the actual cable run length is within spec before quoting.
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- Power source unconfirmed for NHD-600-TX: "NetworkHD 600 encoder" (NHD-600-TX) has no power connection in the schematic. Confirm whether it is PoE-powered, has an included PSU, or requires a separate power supply.
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- Power source unconfirmed for NHD-600-RX: "NetworkHD 600 decoder" (NHD-600-RX) has no power connection in the schematic. Confirm whether it is PoE-powered, has an included PSU, or requires a separate power supply.
- Power source unconfirmed for NHD-600-RX: "NetworkHD 600 decoder" (NHD-600-RX) has no power connection in the schematic. Confirm whether it is PoE-powered, has an included PSU, or requires a separate power supply.

- Power source unconfirmed for NHD-600-RX: "NetworkHD 600 decoder" (NHD-600-RX) has no power connection in the schematic. Confirm whether it is PoE-powered, has an included PSU, or requires a separate power supply.
- Power source unconfirmed for NHD-600-RX: "NetworkHD 600 decoder" (NHD-600-RX) has no power connection in the schematic. Confirm whether it is PoE-powered, has an included PSU, or requires a separate power supply.
- Power source unconfirmed for NHD-CTL-PRO-V2: "NetworkHD controller" (NHD-CTL-PRO-V2) has no power connection in the schematic. Confirm whether it is PoE-powered, has an included PSU, or requires a separate power supply.

This diagram is derived directly from the equipment schedule below - it shows the assumed signal flow from source to display so the customer and installer share the same picture of how the room connects together. The cable schedule underneath lists the transport type, connectors, maximum length, and validation status for each connection.

CABLE RUN	TRANSPORT	MAX LENGTH	CONNECTORS	STATUS	VALIDATION REMINDER
Source 1 (confirm) → NHD-600-TX	Video	15m	SOURCE PORT TBD to HDMI IN	Needs site confirmation	Confirm source and display connector types, cable length (typically <5m passive), HDCP version, EDID behaviour, and resolution/HDR requirements.
Source 2 (confirm) → NHD-600-TX	Video	15m	SOURCE PORT TBD to HDMI IN	Needs site confirmation	Confirm source and display connector types, cable length (typically <5m passive), HDCP version, EDID behaviour, and resolution/HDR requirements.
NHD-600-TX → Network switch	Network / AVoIP	Not confirmed	AV LAN (confirm) to SWITCH PORT 1	Needs site confirmation	Confirm switch port availability, VLAN assignment, PoE budget, cable category, and network topology.

CABLE RUN	TRANSPORT	MAX LENGTH	CONNECTORS	STATUS	VALIDATION REMINDER
NHD-600-TX → Network switch	Network / AVoIP	Not confirmed	AV LAN (confirm) to SWITCH PORT 2	Needs site confirmation	Confirm switch port availability, VLAN assignment, PoE budget, cable category, and network topology.
Network switch → NHD-600-RX	Network / AVoIP	Not confirmed	SWITCH PORT 3 to AV LAN (confirm)	Needs site confirmation	Confirm switch port availability, VLAN assignment, PoE budget, cable category, and network topology.
Network switch → NHD-600-RX	Network / AVoIP	Not confirmed	SWITCH PORT 4 to AV LAN (confirm)	Needs site confirmation	Confirm switch port availability, VLAN assignment, PoE budget, cable category, and network topology.
Network switch → NHD-600-RX	Network / AVoIP	Not confirmed	SWITCH PORT 5 to AV LAN (confirm)	Needs site confirmation	Confirm switch port availability, VLAN assignment, PoE budget, cable category, and network topology.
Network switch → NHD-600-RX	Network / AVoIP	Not confirmed	SWITCH PORT 6 to AV LAN (confirm)	Needs site confirmation	Confirm switch port availability, VLAN assignment, PoE budget, cable category, and network topology.
NHD-CTL-PRO-V2 → Network switch	Network / AVoIP	Not confirmed	AV LAN (confirm) to SWITCH PORT 7	Needs site confirmation	Confirm switch port availability, VLAN assignment, PoE budget, cable category, and network topology.

CABLE RUN	TRANSPORT	MAX LENGTH	CONNECTORS	STATUS	VALIDATION REMINDER
NHD-600-RX → Display 1 (confirm)	Video	15m	HDMI OUT to DISPLAY INPUT (confirm)	Needs site confirmation	Confirm source and display connector types, cable length (typically <5m passive), HDCP version, EDID behaviour, and resolution/HDR requirements.
NHD-600-RX → Display 2 (confirm)	Video	15m	HDMI OUT to DISPLAY INPUT (confirm)	Needs site confirmation	Confirm source and display connector types, cable length (typically <5m passive), HDCP version, EDID behaviour, and resolution/HDR requirements.
NHD-600-RX → Display 3 (confirm)	Video	15m	HDMI OUT to DISPLAY INPUT (confirm)	Needs site confirmation	Confirm source and display connector types, cable length (typically <5m passive), HDCP version, EDID behaviour, and resolution/HDR requirements.
NHD-600-RX → Display 4 (confirm)	Video	15m	HDMI OUT to DISPLAY INPUT (confirm)	Needs site confirmation	Confirm source and display connector types, cable length (typically <5m passive), HDCP version, EDID behaviour, and resolution/HDR requirements.
AV network infrastructure	Network / AVoIP	Not confirmed	RJ45 to RJ45	Needs site confirmation	Confirm switch model, port count, PoE budget, VLAN configuration, multicast policy, and uplink capacity.

Products in the Solution

Product photograph unavailable.

NHD-600-TX — NetworkHD 600 encoder
Role in this design: Encode operator sources

Official product link to be confirmed.

Product photograph unavailable.

NHD-600-RX — NetworkHD 600 decoder
Role in this design: Decode display feeds

Official product link to be confirmed.

Product photograph unavailable.

NHD-CTL-PRO-V2 — NetworkHD controller
Role in this design: Routing control

Official product link to be confirmed.

Complete System Design and Responsibilities

The following items complete the wider solution. Unfilled entries are open design decisions, not supplied equipment or agreed services. Replace each [complete] with the agreed detail or mark it not applicable / excluded.

- Displays / projection | Model, size, location and mounting: [complete] | Supplier / owner: [complete]
- Sources and conferencing | Source devices, meeting platform and USB host: [complete] | Supplier / owner: [complete]
- Audio | Microphones, loudspeakers, DSP and acoustic requirements: [complete] | Supplier / owner: [complete]
- Network | Switches, port capacity, PoE budget, VLANs and IT approval: [complete] | Supplier / owner: [complete]
- Control | User interface, named room modes and programming: [complete] | Supplier / owner: [complete]
- Site infrastructure | Cable routes / distances, rack, ventilation, power and containment: [complete] | Supplier / owner: [complete]
- Delivery and acceptance | Installation, training, support, acceptance tests and dates: [complete] | Supplier / owner: [complete]

Equipment Schedule

ITEM	SKU	DESCRIPTION	ROLE	QTY	TYPE	NOTES
1	NHD-600-TX	NetworkHD 600 encoder	Encode operator	2	Required	Install in the equipment rack.

ITEM	SKU	DESCRIPTION	ROLE	QTY	TYPE	NOTES
			sources			
2	NHD-600-RX	NetworkHD 600 decoder	Decode display feeds	4	Required	One decoder per display.
3	NHD-CTL-PRO-V2	NetworkHD controller	Routing control	1	Required	Confirm firmware compatibility.

Evidence basis: NHD-600-TX: Two confirmed operator sources | NHD-600-RX: Four independently routed displays | NHD-CTL-PRO-V2: Central routing and control requirement

Power Strategy

The power budget below is derived from the governed technical profiles for each selected product. Figures are stated-maximum consumption per the published profile and must be verified against the current datasheet before order.

SKU	QTY	PER-UNIT MAX	TOTAL MAX	POWER DATA (GOVERNED PROFILE)
NHD-600-TX	2	Not proven	—	No power data in governed profile
NHD-600-RX	4	Not proven	—	No power data in governed profile
NHD-CTL-PRO-V2	1	15.4W	15.4W	12V DC | PoE compliant; Max 15.4W, average 8W

Stated maximum consumption: approximately 15W across 1 product with proven figures. 1 product (NHD-CTL-PRO-V2) requires PoE/PoH — confirm the injector or switch PoE budget covers the total.

Options

- No optional enhancements are currently selected.

Assumptions and Exclusions

Assumptions

- **Assumption:** Customer IT will provide a dedicated 10GbE AV VLAN.

Exclusions

- Display, projector, LED and mounting hardware supply unless listed as a specific line item above.
- Electrical, structural and builder's work.
- Network switch procurement, VLAN/multicast configuration and IT commissioning access.

- Structured cabling installation and certification.
- Third-party control programming beyond WyreStorm-native presets.
- Installation, labour and commissioning, unless separately quoted.

Risks and Dependencies

Risks / information required

- **Risk:** Confirm redundant power and switch failover behaviour during commissioning.

Governed dependencies

- **Dependency: NET-SWITCH-10G-MANAGED - Managed 10GbE network switch** (Prompt): Confirm switch model, port budget, VLAN and multicast configuration. Basis: NetworkHD endpoints require managed 10GbE switching.

Implementation and Next Steps

- Approve the network design and complete acceptance testing.

> This proposal draft includes selected WyreStorm products. Validate final product specifications, accessories and dependencies before issue.

> Competitor products are excluded from this proposal and equipment schedule unless a comparison-only output is explicitly requested.

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This proposal has been prepared on a best-efforts basis using WyreStorm Wingman to support your sales and design process. The information contained herein is provided for planning purposes only and does not constitute a binding quotation, order confirmation, or statement of capability.

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Technical Validation: System designs, power budgets, cable schedules, and equipment selections shown in this proposal are based on available product data and assumptions stated within. Final designs must be validated by a qualified AV integrator or pre-sales engineer before installation.

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Confirmed Requirement

Route two operator sources to four displays with resilient 4K distribution.

Design Assumptions

- Customer IT will provide a dedicated 10GbE AV VLAN.

Recommended Architecture

NetworkHD is the current leading product-family direction.

The confirmed routing and resilience requirement needs a scalable 10GbE fabric.

Required WyreStorm Products

- **NHD-600-TX** - NetworkHD 600 encoder
- **NHD-600-RX** - NetworkHD 600 decoder
- **NHD-CTL-PRO-V2** - NetworkHD controller

Optional Enhancements

- No optional enhancements are currently selected.

Risks / Needs Validation

- Confirm redundant power and switch failover behaviour during commissioning.

Next Steps

- Approve the network design and complete acceptance testing.